Unit 1 Progress Check: MCQ Part C

1. Consider the following information about the calculateBill method.

- The method is a static method in the Store class.
- The method has one int parameter.
- The method has return type double.
- The method can be accessed from another class.

Which of the following code segments, when appearing in a method in a class other than Store, will compile without error?

(A)

```
double tot = calculateBill(5);
```

(B) ✓

```
double tot = Store.calculateBill(5);
```

(C)

```
double tot;

tot.calculateBill(5);
```

(D)

```
double tot = 5.0;

tot.calculateBill();
```

Answer B

Correct. Since the method is static, it is called using the class name Store along with the dot operator. The method call uses a valid int argument (5). The method's return value is assigned to a variable that is the same as the method's return type (double).

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2. Consider the following information about the getTax method.

- The method is a static method in the PurchaseManager class.
- The method has one double parameter.
- The method returns a double equal to its parameter times 0.08.

The following code segment appears in another method in the PurchaseManager class.

```
double tax = getTax(100.0);
System.out.println(tax);
```

What, if anything, is printed as a result of executing the code segment?

(A) 8.0 ✓

(B) 108.0

(C) Nothing is printed because the method call should be PurchaseManager.getTax(100.0).

(D) Nothing is printed because the return value from getTax cannot be assigned to the variable tax.

Answer A

Correct. The method returns its parameter 100.0 times 0.08, which is 8.0. Since the call to this static method appears in the same class, the class name PurchaseManager is not needed in the call.

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3. Consider the following information about the generatePasscode method.

- The method appears in the AccountManager class.
- The method has no parameters.
- The method has return type int.
- The method can be accessed from another class.

The following code segment appears in a method in a class other than AccountManager class.

```
int c = AccountManager.generatePasscode();
System.out.println(c);
```

Which of the following must be true for the code segment to compile without error?

(A) The generatePasscode method should be designated as void.

(B) The generatePasscode method should be designated as static. ✓

(C) The AccountManager class should contain another method with the name generatePasscode that has at least one parameter.

(D) The AccountManager class should contain another method with the name generatePasscode that has return type double.

Answer B

Correct. Since the method call uses the class name AccountManager along with the dot operator, it is associated with the class, not instances of the class. Class methods are declared using the keyword static.

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4. Consider the following statement, which is intended to generate an integer between 1 and 10, inclusive, and assign it to result. This statement does not work as intended.

```
int result = (int) Math.random() * 10 + 1;
```

Which of the following changes can be made so that this statement works as intended?

(A) Add parentheses around the expression 10 + 1, as shown in the following statement.

```
int result = (int) Math.random() * (10 + 1);
```

(B) Add parentheses around the expression Math.random() * 10, as shown in the following statement. ✓

```
int result = (int) (Math.random() * 10) + 1;
```

(C) Change the value of 10 to 9, as shown in the following statement.

```
int result = (int) Math.random() * 9 + 1;
```

(D) Remove the cast of Math.random() to an int value, as shown in the following statement.

```
int result = Math.random() * 10 + 1;
```

Answer B

Correct. Adding parentheses around the multiplication ensures that the cast to an int value occurs after the result of the call to Math.random() has been scaled to an appropriate size.

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5. Consider the following code segment.

```
double result = Math.abs(x);
```

Which of the following statements about the variable x is true?

(A) x may hold any numerical value. ✓

(B) x must be equal to 0.

(C) x must have a value greater than 0.

(D) x must have a value less than 0.

Answer A

Correct. Any numerical int or double value may be used here.

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6. Consider the following code segment.

```
int num = (int)(Math.random() * 6);
System.out.println(num);
```

Which of the following **cannot** be printed as a result of executing this code segment?

(A)

0

(B)

2

(C)

4

(D) ✓

6

Answer D

Correct. The given expression returns an integer greater than or equal to 0, but less than 6. The value 6 is not within this range.

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7. A teacher has created a Student class. The class contains the following.

- An int variable named grade to represent the student's grade level
- A String variable named name to represent the student's name
- A double variable named average to represent the student's grade point average
- A method named updateAverage that updates the value of average.

The object greg will be declared as type Student.

Which of the following descriptions is accurate?

(A) greg is an instance of the Student class. ✓

(B) greg is an instance of three attributes.

(C) Student is an instance of the greg object.

(D) updateAverage is an instance of the Student class.

Answer A

Correct. The object greg has been declared as type Student, which creates it as an instance of the Student class.

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8. A school administrator has created a Student class. The class contains variables to represent the following.

- An int variable named studentID to represent the student's ID number
- A String variable named studentName to represent the student's name

The school administrator has also created a Parent class. The class contains variables to represent the following.

- A String variable named parentName to represent the parent's name
- A String variable named email to represent the parent's e-mail address

The object penelope will be declared as type Student. The object mrsPatel will be declared as type Parent.

Which of the following descriptions is accurate?

(A) An attribute of the penelope object is email.

(B) An attribute of the penelope object is Parent.

(C) An attribute of the penelope object is Student.

(D) An attribute of the mrsPatel object is email. ✓

Answer D

Correct. The mrsPatel object is an instance of the Parent class. The mrsPatel object has two attributes: parentName and email.

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9. The Dog class has a constructor that takes two parameters. The first parameter is a String representing a dog's name and the second parameter is an int representing the dog's age, in months.

The following code segment appears in a class other than Dog.

```
Dog dog1 = new Dog("Rex", 4);
Dog dog2 = dog1;
dog1 = new Dog("Fido", 60);
```

Which of the following best describes the contents of dog1 and dog2 as a result of executing this code segment?

(A) dog1 and dog2 are Dog variables that each contain the same Dog object, which represents a dog named "Fido" that is 60 months old.

(B) dog1 is a Dog variable that contains a Dog object representing a dog named "Fido" that is 60 months old, and dog2 is a Dog variable that contains a Dog object representing a dog named "Rex" that is 4 months old.

(C) dog1 and dog2 are Dog reference variables that each contain a reference to the same memory address that contains the Dog object representing a dog named "Fido" that is 60 months old.

(D) dog1 is a Dog reference variable, which contains a reference to the memory address for the Dog object representing a dog named "Fido" that is 60 months old. dog2 is a Dog reference variable, which contains a reference to the memory address for the Dog object representing a dog named "Rex" that is 4 months old. ✓

Answer D

Correct. Object reference variables store memory locations where objects are stored. The dog1 variable is assigned a new memory location in the third assignment statement.

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10. The Vbox class includes the following two constructors.

Constructor Signature	Explanation
public vBox(int w, int h, int d)	Constructs a Vbox object that represents a box with width w, height h, and depth d.
public vBox(int len)	Constructs a Vbox object that represents a box with width len, height len, and depth len.

Which of the following declarations, appearing in a class other than Vbox, will correctly instantiate a Vbox object?

(A)

```
Vbox b1 = new Vbox(4.0);
```

(B)

```
Vbox b2 = new Vbox(4, 2);
```

(C) ✓

```
Vbox b3 = new Vbox(2, 8, 4);
```

(D)

```
Vbox b4 = new Vbox(4.0, 4.0, 4.0);
```

Answer C

Correct. This declaration creates a Vbox object using the constructor with the signature Vbox(int w, int h, int d), which has three int parameters.

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11. The Person class has exactly two constructors. Partial declarations of the constructors are shown.

```
public Person(int idNumber, boolean isActive)
{ /* implementation not shown */ }

public Person(int idNumber, int age, boolean isActive)
{ /* implementation not shown */ }
```

Which of the following statements does **not** correctly create an object of type Person?

(A)

```
Person p1 = new Person(1138, true);
```

(B)

```
Person p2 = new Person(3136, 17, false);
```

(C) ✓

```
Person p3 = new Person(3544, false, 18);
```

(D)

```
Person p4 = new Person(0, false);
```

Answer C

Correct. This statement attempts to use the second constructor. The list of arguments given is not in the correct order by type. The second argument should be an int and the third argument should be a boolean.

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12. The Player class has only one constructor. A partial declaration of the constructor is shown.

```
public Player(boolean isOnline, int numLives)
{ /* implementation not shown */ }
```

Which of the following statements correctly creates a Player object?

(A) ✓

```
Player p = new Player(true, 5);
```

(B)

```
Player p = new Player(5, true);
```

(C)

```
Player p = Player(true, 5);
```

(D)

```
Player p = Player(5, true);
```

Answer A

Correct. This is a correct call to instantiate an object of type Player. The call uses the keyword new and uses the name of the class as the type. The call has the correct number of parameters, the parameter types are correct, and the parameters are in the correct order.

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13. The AnimalPrinter class includes the following two methods.

Method Signature	Explanation
public void printDog()	Prints the word "dog" and then moves the cursor to a new line.
public void printCat()	Prints the word "cat" and then moves the cursor to a new line.

The method myMethod appears in a class other than AnimalPrinter. The method is intended to produce the following output.

```
dog
cat
```

Assume that an AnimalPrinter object myPrinter has been properly declared and initialized inside myMethod. Which of the following code segments, if located in myMethod, will produce the intended output?

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(A)

```
printDog();

printCat();
```

(B)

```
printDog(AnimalPrinter);

printCat(AnimalPrinter);
```

(C)

```
AnimalPrinter.printDog();

AnimalPrinter.printCat();
```

(D) ✓

```
myPrinter.printDog();

myPrinter.printCat();
```

Answer D

Correct. To work correctly, the dot operator must be used to invoke a non-static method of a class on an object of that class. This code segment correctly invokes the myPrinter object's methods printDog() and printCat() and will produce the desired output.

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14. Consider the following information about the Person class.

- The class has an int attribute age that represents a person's age.
- The class has a non-static getAge method that has no parameters and returns the value of age. This method can be called from another class.

The following code segments each appear in a class other than Person. Assume that p is a Person object whose age attribute is equal to 16.

Code Segment 1

```
int val1 = p.getAge();
System.out.println(val1);
```

Code Segment 2

```
int val2 = Person.getAge();
System.out.println(val2);
```

What, if anything, is printed as a result of executing each of the code segments?

(A) Both code segments print 16.

(B) Code segment 1 prints 16, but code segment 2 prints nothing because it will not compile. ✓

(C) Code segment 2 prints 16, but code segment 1 prints nothing because it will not compile.

(D) Neither code segment will print anything because they will not compile.

Answer B

Correct. Method getAge is non-static, which means it is called on objects of the Person class along with the dot operator. It cannot be called on the Person class itself.

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15. The Dog class contains a non-static method named calculateWalkLength which can be called from a class other than Dog. The method takes an integer parameter for the number of minutes a dog is taken for a walk and returns a double representing the number of miles walked during that time.

The following code segment appears in a class other than Dog. The statement in line 2 causes a compile-time error.

```
Dog fido = new Dog();                        // line 1
double miles = calculateWalkLength(fido, 90); // line 2
```

Which of the following can be used to replace line 2 so that this code segment will compile without error?

(A)

```
double miles = calculateWalkLength(90);
```

(B)

```
double miles = Dog.calculateWalkLength(fido, 90);
```

(C)

```
double miles = Dog.calculateWalkLength(90);
```

(D) ✓

```
double miles = fido.calculateWalkLength(90);
```

Answer D

Correct. The method is called with an object of the Dog class and it accepts one integer parameter.

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16. Consider the following code segment.

```
String one = "computer";
String two = "science";
String concat = /* missing code */;
System.out.println(concat);
```

Which of the following expressions can be used to replace `/* missing code */` so that the code segment prints the string `"pun"`?

(A)

```
one.substring(3, 4) + two.substring(4, 4)
```

(B) ✓

```
one.substring(3, 5) + two.substring(4, 5)
```

(C)

```
one.substring(4, 5) + two.substring(5, 5)
```

(D)

```
one.substring(4, 6) + two.substring(5, 6)
```

Answer B

Correct. The call `one.substring(3, 5)` returns `"pu"`, and the call `two.substring(4, 5)` returns `"n"`. The concatenated result is `"pun"`.

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17. Consider the following code segment, which is intended to create a String that consists of the last character in word and assign it to the variable lastChar.

```
String word = /* initialization not shown */;  // line 1
int len = word.length();                        // line 2
String lastChar = word.substring(len);          // line 3
```

Which of the following best describes why this code segment will not work as intended?

(A) substring needs to be called with two parameters, so the method call in line 3 should be changed to word.substring(len, len + 1).

(B) len is not the correct index of the last character in word, so the method call in line 3 should be changed to word.substring(len - 1). ✓

(C) len is not the correct index of the last character in word, so the method call in line 3 should be changed to word.substring(len + 1).

(D) substring cannot be called with a variable as the parameter, so the method call in line 3 should be changed to word.substring(word.length()).

Answer B

Correct. Since String objects are indexed starting at 0, the last character will be located at index word.length() – 1.

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18. The following code segment is intended to assign to newWord the result created by removing the first occurrence of "a" from word. Assume that the String variable word has been properly declared and initialized. This code segment works for some, but not all, values of word.

```
int aLoc = word.indexOf("a");
String newWord = word.substring(0, aLoc) + word.substring(aLoc + 1);
```

Which of the following conditions best describes the condition in which this code segment will not work as intended and will result in a runtime error?

(A) "a" is the first character in word.

(B) "a" is the last character in word.

(C) word does not contain the character "a". ✓

(D) word contains only a sequence of multiple "a" characters.

Answer C

Correct. indexOf returns -1 when the target String cannot be found. Using -1 as a parameter in substring will result in a StringIndexOutOfBoundsException error.